Transparent Saliency Routing for Temporal Image Drift

A compact mock research paper • fixed-horizon prototype run • 29 September 2026

Abstract

This prototype evaluates a ViT-only drift correction pipeline whose update decision can be inspected directly. Historical models train on 2004–2008, model selection uses 2009, drift adaptation uses 2010, router calibration uses 2011, and 2012–2014 remain untouched until final evaluation. Under combined drift in 2014, the transparent router improves macro F1 by 1.16 percentage points over the historical XAI model and retains 82.3% of the gain available from always choosing the updated model.

1. Research question

Can visible saliency change and prediction evidence determine when a drift-adapted ViT should replace a historical ViT prediction across a realistic future horizon?

2. Architecture and transparent routing

The experiment compares a baseline ViT, an attention-regularised XAI ViT, and a paired historical/newer XAI system. Its router is a standardized logistic rule over four logged quantities: saliency Jensen–Shannon divergence, confidence gain, entropy reduction, and class-evidence gain. A minimum saliency-change floor must also be passed. The saved coefficients and threshold make every update decision reproducible.

3. Chronological protocol

The split models deployment: historical learning stops before adaptation, and the future horizon is used only for final measurement. This prevents year-by-year retraining from leaking test information into the model.

Stage	Years	Purpose
Historical training	2004–08	Fit original ViTs
Selection	2009	Choose checkpoints
Adaptation	2010	Update XAI ViT with mild drift
Calibration	2011	Fit explicit route rule
Frozen test	2012–14	Measure future drift

3.1 Optional realistic drift

The drift toggle applies mild saturation change, Gaussian sensor noise, and partial occlusion. The newer ViT sees a deterministic 50/50 mixture of clean and individually corrupted 2010 samples. The router is calibrated on moderate individual corruptions in 2011. Combined corruption is withheld until 2012–2014, making it a composition shift rather than a memorised augmentation recipe.

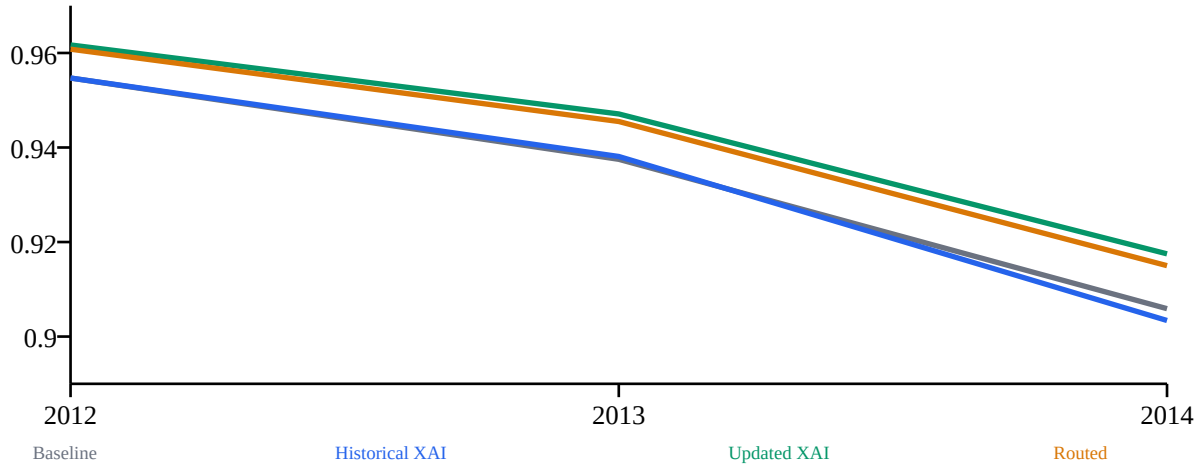
3.2 Saved routing policy

Transparent feature	Coefficient
Saliency Js	+0.240
Confidence Gain	+0.440
Entropy Reduction	-0.321
Evidence Gain	-0.055
Route probability threshold	0.35
Saliency-change floor	0.0022

Coefficients act on standardized features. Their signs describe the 2011 calibration relationship; they are not causal effects. The explicit saliency floor ensures that saliency remains part of the decision rather than a decorative post-hoc plot.

4. Fixed-horizon results

Macro F1 under combined saturation, noise, and occlusion. The widening decline across years makes the temporal effect visible.



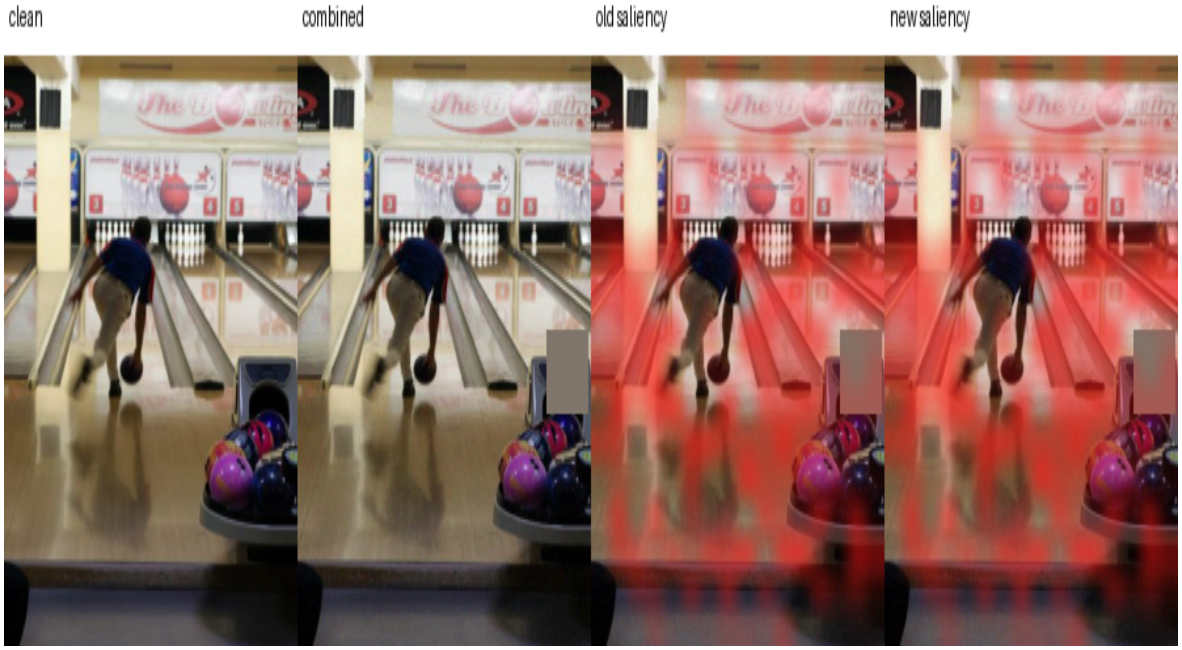
Year	Baseline	Hist. XAI	Updated XAI	Routed	Helpful / harmful
2012	95.47	95.47	96.17	96.08	45 / 14
2013	93.75	93.81	94.71	94.55	72 / 32
2014	90.59	90.34	91.75	91.50	120 / 48

In 2014, the updated XAI model gains 1.41 points over historical XAI. Routing retains 1.16 points: (91.50–90.34)/(91.75–90.34) = 82.3% of the available gain. Clean-data macro F1 is 95.13% baseline, 95.24% historical XAI, 95.85% updated XAI, and 95.46% routed.

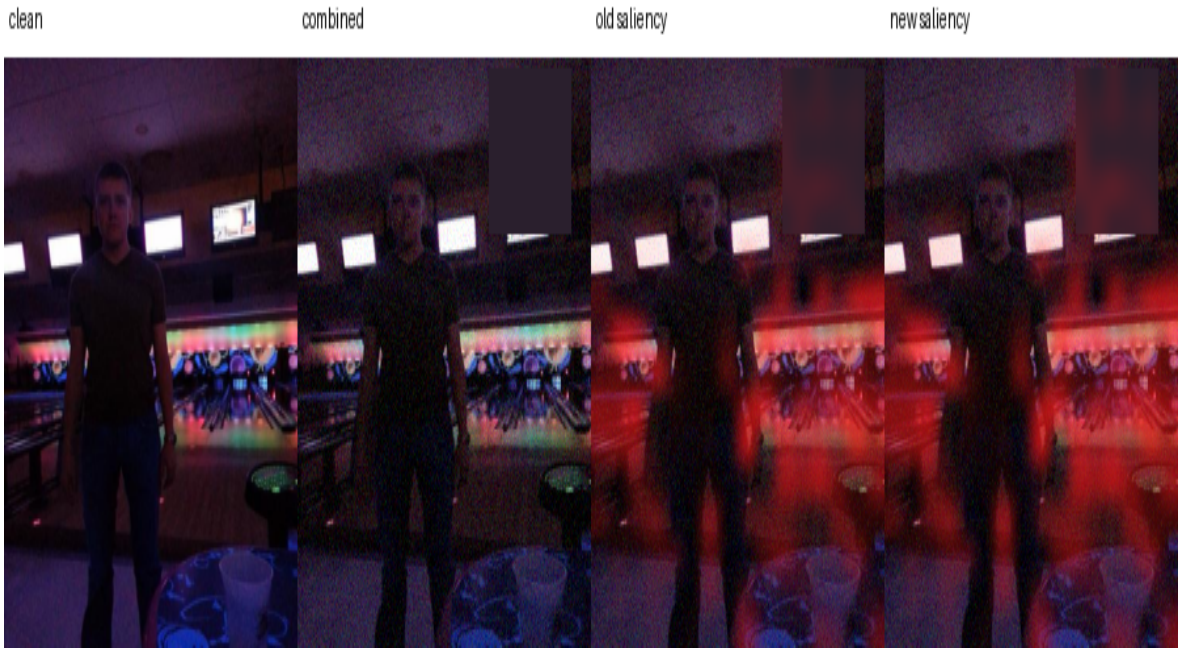
5. Saliency audit examples

Each panel shows the clean reference, drifted input, historical saliency, and updated saliency. Together with the saved decision log, this lets a reviewer connect the route to visible attention changes and prediction evidence.

Combined drift • 2012



Combined drift • 2014



6. Interpretation

Compound drift produces a clearer gap than the earlier one-year test. The adapted ViT improves both drifted and clean predictions. The router captures most of that improvement while exposing its decision variables. Confidence gain has the strongest positive learned coefficient; saliency divergence contributes positively and is reinforced by the explicit saliency floor. Other coefficient signs describe correlations in the 2011 calibration set and should not be treated as causal explanations.

7. Limitations and next experiment

This is one exploratory run with one seed, synthetic corruptions, and one calibrated threshold. It does not provide confidence intervals or an external validation dataset. A defensible final study should repeat the complete chronological protocol across seeds, bootstrap the year-level differences, and report always-old, always-new, confidence-only, and saliency-only ablations. A second temporally indexed dataset would test whether the route rule transfers beyond this stream.

8. Reproducibility

The repository includes a compact JSON summary, the learned router policy, 45 saliency panels, and the full source used to run the fixed horizon. Large checkpoints remain on the server. The stored run identifiers are baseline 20260928T152657Z-horizon, historical XAI 20260928T161159Z-horizon-xai, and router 20260928T161159Z-horizon-router.

References

[1] Dosovitskiy et al., Vision Transformer, 2020. [2] Oquab et al., DINOv2, 2023. [3] Hendrycks & Dietterich, ImageNet-C, 2019. [4] Hendrycks et al., AugMix, 2019. [5] Lin et al., CLEAR, 2021. Full links appear in the Markdown version.